

Integral Representation of Zeta(2)

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1 Integral Representation of Zeta(2)

$$\int_0^\infty \frac{t}{e^{\alpha t} - 1} dt = \int_0^\infty \frac{te^{-\alpha t}}{1 - e^{-\alpha t}} dt = \int_0^\infty t \sum_{n=1}^\infty e^{-n\alpha t} dt$$

By the Dominant Convergence Theorem: (Include Reference)

$$\int_0^\infty t \sum_{n=1}^\infty e^{-n\alpha t} dt = \sum_{n=1}^\infty \int_0^\infty te^{-n\alpha t} dt$$

Using $n \cdot \alpha \cdot t = x \rightarrow n \cdot \alpha \cdot dt = dx$

$$\sum_{n=1}^\infty \int_0^\infty \frac{x}{n\alpha} e^{-x} \frac{dx}{n\alpha} = \sum_{n=1}^\infty \frac{1}{n^2 \alpha^2} \int_0^\infty x e^{-x} dx = \frac{1}{\alpha^2} \sum_{n=1}^\infty \frac{1}{n^2} \Gamma(2)$$

Recall $\Gamma(2) = 1! = 1$ (Include Reference)

$$\frac{1}{\alpha^2} \sum_{n=1}^\infty \frac{1}{n^2} \Gamma(2) = \frac{1}{\alpha^2} \sum_{n=1}^\infty \frac{1}{n^2} = \frac{1}{\alpha^2} \zeta(2)$$

Recall $\zeta(2) = \frac{\pi^2}{6}$ (Include Reference)

$$\Rightarrow \int_0^\infty \frac{t}{e^{\alpha t} - 1} dt = \frac{\pi^2}{6\alpha^2}$$